INCH-POUND

MIL-S-20708/16F
13 March 1989

SUPERSEDING
MIL-S-20708/16E
30 July 1987

# MILITARY SPECIFICATION SHEET
# SYNCHRO, CONTROL DIFFERENTIAL TRANSMITTER, TYPE 15CDX4D

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708

TABLE I Requirements

| Requirement | Value | Unit | Tolerance |
|---|---|---|---|
| Frequency | 400 | Hz | ±1% |
| Primary Voltage | 78 Ⓕ | volts | ±1% |
| Primary Current | 90 0 | milliamps | maximum |
| Primary Power | 1 1 | watts | maximum |
| Impedance | | | |
| Zso | 870-1150 | ohms | min -max |
| Zrs | 195-290 | ohms | min -max |
| Impedance Angle | | | |
| Zso | 79 0-85 0 | degrees | min -max |
| Zrs | 27 0-37 0 | degrees | min -max |
| Transformation Ratio | 1 154 | ------ | ±2% |
| Phase Shift (Lead) | 5 5 | degrees | ±2 0 |
| Electrical Error | 6 0 | minutes | maximum |
| Null Voltage | | | |
| Total | 60 0 | millivolts | maximum |
| Fundamental | 32 0 | millivolts | maximum |
| Friction Torque | 0 07 | ounce-inches | maximum |
| Radial Play | 0 0006 | inches | maximum |
| End Play | 0 0010 | inches | ±0 0005 |
| Temperature Rise | 20 0 | degrees (C) | maximum |

Ⓕ denotes changes


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TABLE II Military part number variant characteristics

| Military Part No 1/ | A ± 010 | B ± 010 | G Maximum | Shaft Style | Terminal End | U 2/ + 0000 - 0002 |
|---|---|---|---|---|---|---|
| M20708/16-01D | 555 | 500 | 1 798 | Splined | Terminal Board | 1872 |
| M20708/16-02D | ↓ | ↓ | 1 798 | Plain & Slotted | Terminal Board | ↓ |
| M20708/16-03D | ↓ | ↓ | 1 536 | Plain & Slotted | Wire Leads | ↓ |

(F)

1/Military part number M20708/16-01D shall be the NATO standard

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation

(F) 2/The tolerance for plain and slotted shaft diameters is + 0000 and - 0005

TABLE III Military part number cross-references

| SUPERSEDED MILITARY PART NUMBERS | | NEW MILITARY PART NUMBERS |
|---|---|---|
| MIL-S-20708/16D | MIL-S-20708/16C | |
| M20708/16D-001 | M20708/16-001 | M20708/16-01D |
| M20708/16D-002 | | M20708/16-02D |
| M20708/16D-003 | | M20708/16-03D |

NOTE All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification Part number changes do not affect form, fit or function of the synchros listed therein

Custodians
Army-AR
Navy-AS
Air Force-85

Review Activities
Army-MI,AV
Navy-OS
Air Force-11,17,99
DLA-ES

User Activities
Navy-MC,CG

Preparing Activity
Navy-AS

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